Exploring the Potential of Data Analysis and Machine Learning in the Beer Industry: A Study of IPA Styles

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1. Project Overview

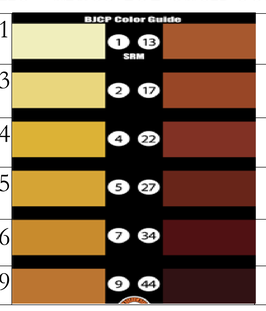
The goals for this project include creating a model that could identify Black IPA's and the implementation of R Shiny as a functional database for beer information. Metrics and Terminology come from the Beer Judge Certification Program's website.

2. Beer Terminology

Metrics

- ABV: Alcohol by Volume is a standard measure of how much alcohol is contained given the volume of an alcoholic beverage
- IBU: International Bitterness Units measure the parts per million of isohumulone from hops in a beer
- Color: Beer color is measured on a scale from light to dark using the SRM. This table from the BJCP shows numeric values for each color classification

Light Straw:	1	Copper:	10-17
Straw:	2-3	Light Brown:	17-18
Yellow:	3-4	Brown:	19-22
Light Gold	4-5	Dark Brown:	22-30
Gold:	5-6	Black:	30+
Amber:	6-9	Black, Opaque:	40+



Styles/Terms

- Beer Style: Beer styles are defined by set characteristics including flavor profile, color, alcohol by volume, as well as any special techniques and ingredients used
  - Black IPA: "A beer with the dryness, hop-forward balance, and flavor characteristics of an American IPA, only darker in color."
  - Juicy: "a trendy term used to describe hops that have a quality like fresh fruit juices, especially tropical fruits."

3. Logistic Regression

	Coef.	S.E.	Z-val	P-val
ABV	-0.188	0.147	-1.280	0.200
IBU	-0.001	0.003	-0.375	0.707
Color	0.229	0.013	17.479	<2e-16

Goal 1: Identify Black IPA's

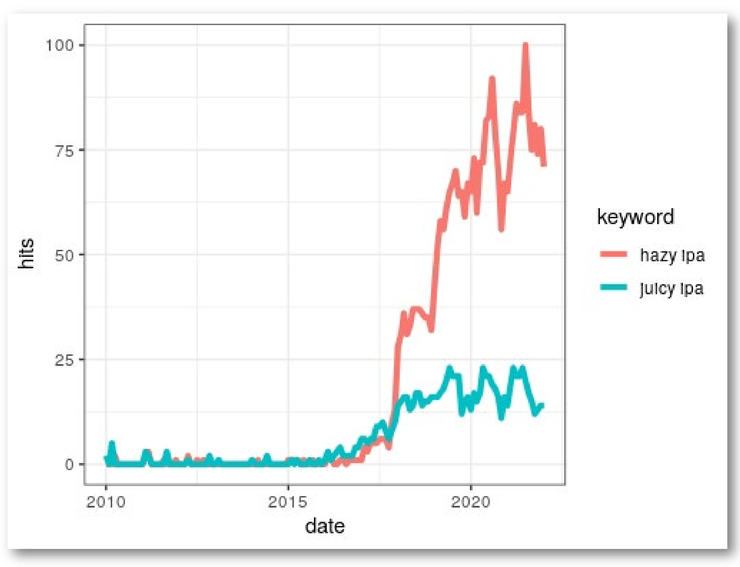
To train a logistic regression model to identify Black IPA's even if they are not labeled as such.

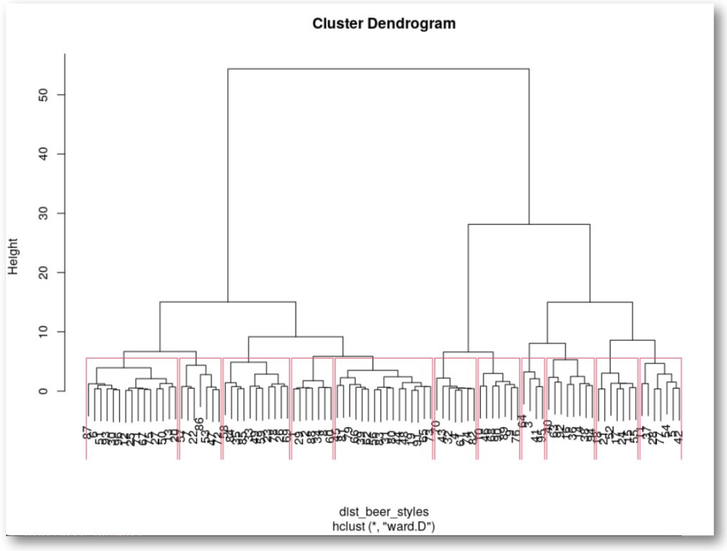
Color has the most impact on the model, agreeing with the BJCP definition of a Black IPA, "only darker in color". Because Color was the only significant variable, the model had a difficult time identifying IPA's that might qualify as Black IPA's but were not labeled as such. IPA styles with similar colors had labels such as "Brown", "Chocolate", "Dark", or even "Red". The data used for training and testing was very imbalanced because there was only a small percentage of labeled Black IPA's.

4. Shiny Application

Goal 2: Visualize IPA Trends

Implement a Shiny app that tracks the number of searches for a term over time. Using Google Trends, I created a plot to search for the number of "hits" of the names of certain IPA styles.





Goal 3: Beer Recommendation

Use a hierarchal clustering system to group similar commercial beers with each other to recommend.

The app takes in a commercial beer that the user selects, finds its style and which cluster it belongs to, then returns a list of similar commercial beers a user might like.